



Benefit from public unlabeled data: A Frangi filter-based pretraining network for 3D cerebrovascular segmentation

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ABSTRACT

Precise cerebrovascular segmentation in Time-of-Flight Magnetic Resonance Angiography (TOF-MRA) data is crucial for computer-aided clinical diagnosis. The sparse distribution of cerebrovascular structures within TOF-MRA images often results in high costs for manual data labeling. Leveraging unlabeled TOF-MRA data can significantly enhance model performance. In this study, we have constructed the largest preprocessed unlabeled TOF-MRA dataset to date, comprising 1510 subjects. Additionally, we provide manually annotated segmentation masks for 113 subjects based on existing external image datasets to facilitate evaluation. We propose a simple yet effective pretraining strategy utilizing the Frangi filter, known for its capability to enhance vessel-like structures, to optimize the use of the unlabeled data for 3D cerebrovascular segmentation. This involves a Frangi filter-based preprocessing workflow tailored for large-scale unlabeled datasets and a multi-task pretraining strategy to efficiently utilize the preprocessed data. This approach ensures maximal extraction of useful knowledge from the unlabeled data. The efficacy of the pretrained model is assessed across four cerebrovascular segmentation datasets, where it demonstrates superior performance, improving the cDice metric by approximately 2%–3% compared to the latest semi- and self-supervised methods. Additionally, ablation studies validate the generalizability and effectiveness of our pretraining method across various backbone structures. The code and data have been open source at: <https://github.com/shigen-Stoneroot/FFPN>.

1. Introduction

TOF-MRA is one non-invasive medical imaging technique to visualize cerebral blood vessels. It provides detailed images of the blood vessels in the brain, enabling early detection and treatment of potentially life-threatening conditions (Özsarlak et al., 2004; Hassouna et al., 2006). The accurate vessel segmentation is a crucial preprocessing step in TOF-MRA image analysis, which provides surgeons with essential information about the location, size, and connectivity of blood vessels and facilitates surgical planning and intervention (Ni et al., 2020).

Compared to the segmentation of other biological tissues, brain vessel segmentation based on TOF-MRA presents more challenges. Most organs typically have a spherical shape and are distributed in a concentrated manner, while blood vessels have a tubular structure and their distribution in TOF-MRA is relatively sparse (Chen et al., 2022a; Xia et al., 2022), which increases the difficulty and cost in obtaining manually labeled MRA-TOF data. On the contrary, the unlabeled TOF-MRA data are easily accessible from the public resource (i.e., IXI

and OASIS3). Therefore, how to fully use the public unlabeled data is essential for cerebrovascular segmentation with limited manually labeled data.

There are two principal deep learning (DL)-based methodologies capable of leveraging unlabeled data: semi-supervised learning (SemiSL) (Qi and Luo, 2020; Yang et al., 2022) and self-supervised learning (SSL) (Ericsson et al., 2022; Huang et al., 2023) (1a and b). SemiSL methods typically integrate unlabeled data with labeled data during training, emphasizing the consistency between them (Cheplygina et al., 2019). This integration often necessitates repeated access to the unlabeled data, increasing computational demands, particularly when multiple research institutions are involved. Additionally, public unlabeled data, often sourced from diverse sites, can exhibit high heterogeneity compared to single-site datasets. This heterogeneity introduces distribution shifts that could potentially degrade the performance of SemiSL methods (Chen et al., 2019). Conversely, SSL methods generally begin

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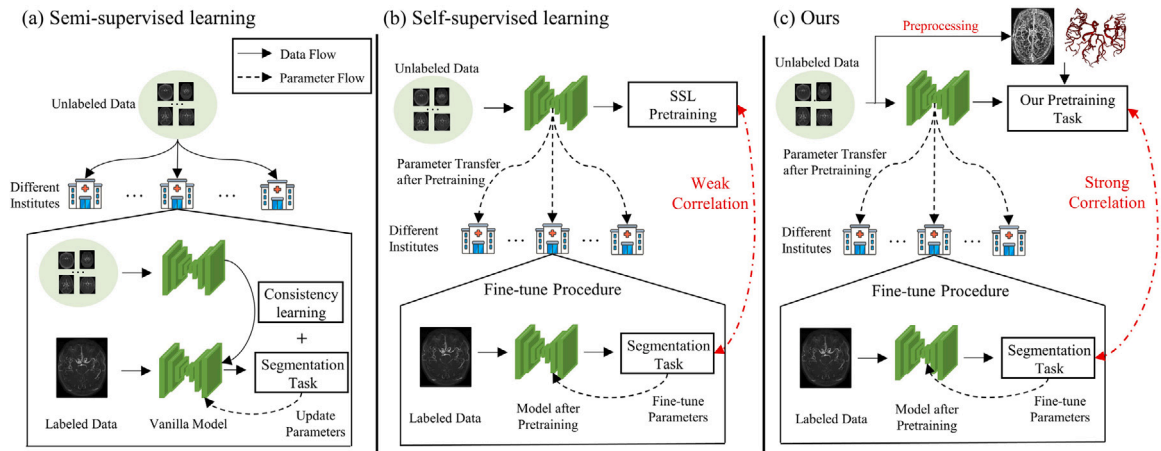


Fig. 1. The illustration of using unlabeled MRA-TOF data through (a) semi-supervised learning (SemiSL), (b) self-supervised learning (SSL), and (c) our pretraining method. In SemiSL, institutes repeatedly access both labeled and unlabeled data. In SSL, each institute can simply use the model parameters after pretraining on the unlabeled data and then fine-tune the model on their labeled data. Our method optimizes the pretraining task (red arrow), enhancing similarity to downstream tasks and improving fine-tuning performance.

by pretraining the model on a pretext task (Zhang et al., 2023). Then the pretrained model can be adapted by different institutions without needing direct access to the unlabeled data thus enhancing patient privacy (Asadian et al., 2022). This approach may potentially decrease the impact of distribution drift by segregating the processes of data usage thereby enhancing the robustness of the model against such variations.

However, the effectiveness of SSL methods is closely linked to the pretext tasks. Current tasks, such as contrastive learning between augmented data (Chen et al., 2020; Liao et al., 2022) and masked image modeling (He et al., 2022), typically diverge from downstream tasks, such as cerebrovascular segmentation in this study. While these SSL settings are beneficial for generalizing across multiple downstream tasks, their effectiveness may be limited for specific, singular tasks. Thus, integrating domain-specific knowledge into the pretraining process could be crucial for enhancing model performance. Additionally, various techniques exist for enhancing and extracting tubular structures in medical imaging, such as the Gabor filter (Aguirre-Ramos et al., 2018), Kirsch transform (Nath et al., 2020), vessel-enhanced diffusion (Manniesing et al., 2006), and the Frangi filter (Frangi et al., 1998). The Frangi filter, in particular, has been extensively validated for its robustness in enhancing 3D vascular structures (Phellan and Forkert, 2017; Ruhaiyem and Hammade, 2021; Jimenez-Carretero et al., 2013; Mahapatra et al., 2022). This filter technique, which analyzes local image intensity and the Hessian matrix to detect vessel-like structures, is fundamental in traditional vessel segmentation algorithms. Therefore, incorporating the Frangi filter into the pretraining could be highly beneficial for tasks such as vessel segmentation.

Here, we introduce the Frangi Filter-Based Pretraining Network (FFPN) to fully utilize publicly available unlabeled TOF-MRA data. We have developed a preprocessing workflow tailored for large-scale unlabeled datasets. To alleviate the computational demands associated with processing 1510 volumetric data, we devised an efficient cropping strategy that significantly reduces image size. Following cropping, the Frangi filter is applied to enhance vascular structures, resulting in vessel enhanced images. The preprocessing sequence also includes thresholding methods and connected component analysis to produce a coarse vessel segmentation. These vessel enhanced images and coarse segmentations are then employed in a pretraining task strongly correlated to the downstream task (Fig. 1c). Our model is pretrained on the large TOF-MRA dataset and is subsequently evaluated across four labeled datasets. The segmentation outcomes demonstrate that FFPN surpasses state-of-the-art SemiSL and SSL baseline models. Additionally, ablation studies validate that our specialized pretraining task not only enhances model performance with fewer labeled datasets but also consistently

boosts performance across various backbone structures.

Our main contributions are summarized as follows:

- We develop an automated preprocessing workflow to efficiently handle unlabeled data. The preprocessing provides the vessel enhanced images and coarse vessel segmentations that are used for the pretraining procedure.
- We propose a simple yet effective pretraining strategy by using the preprocessed TOF-MRA data. This pretraining task contains regression learning, coarse label segmentation, and max intensity projection (MIP) consistency learning. The results show that the proposed method shows great superiority over existing models.
- We curated and organized a large unlabeled dataset (1510 volumetric data) and three manual-annotated datasets (a total of 113 volumetric data) based on existing publicly available image datasets. These data are expected to advance semi-supervised, weakly supervised, and self-supervised learning methods in cerebrovascular segmentation.

2. Related work

2.1. Cerebrovascular segmentation

Traditional approaches to cerebrovascular segmentation have predominantly relied on the analysis of grayscale continuity within images, employing threshold-based segmentation techniques (Otsu, 1979; Frangi et al., 1998). Additionally, statistical modeling-based methods capitalize on the distinct grayscale distributions exhibited by various brain tissues during imaging to enhance segmentation accuracy (Duan et al., 2019; Alderliesten et al., 2006). Furthermore, methods based on the Frangi filter have demonstrated considerable success in delineating cerebrovascular structures in TOF-MRA images (Ruhaiyem and Hammade, 2021; Jerman et al., 2015). In recent years, DL-based techniques have achieved remarkable advancements in vessel segmentation tasks (Sanches et al., 2019; Sichtermann et al., 2019; Qi et al., 2023). Notably, Livne et al. developed a UNet-based model specifically for segmenting vessels in patients with cerebrovascular disease (Livne et al., 2019). Additionally, Chen et al. introduced an attention-assisted generative adversarial network (A-SegAN) that facilitates automatic cerebrovascular segmentation (Chen et al., 2022b). For a comprehensive overview of these developments and others, readers are encouraged to consult the detailed review by Chen et al. (2023).

2.2. SemiSL and SSL methods for medical images

Numerous SemiSL and SSL methods have been developed to leverage unlabeled medical data for enhancing model performance

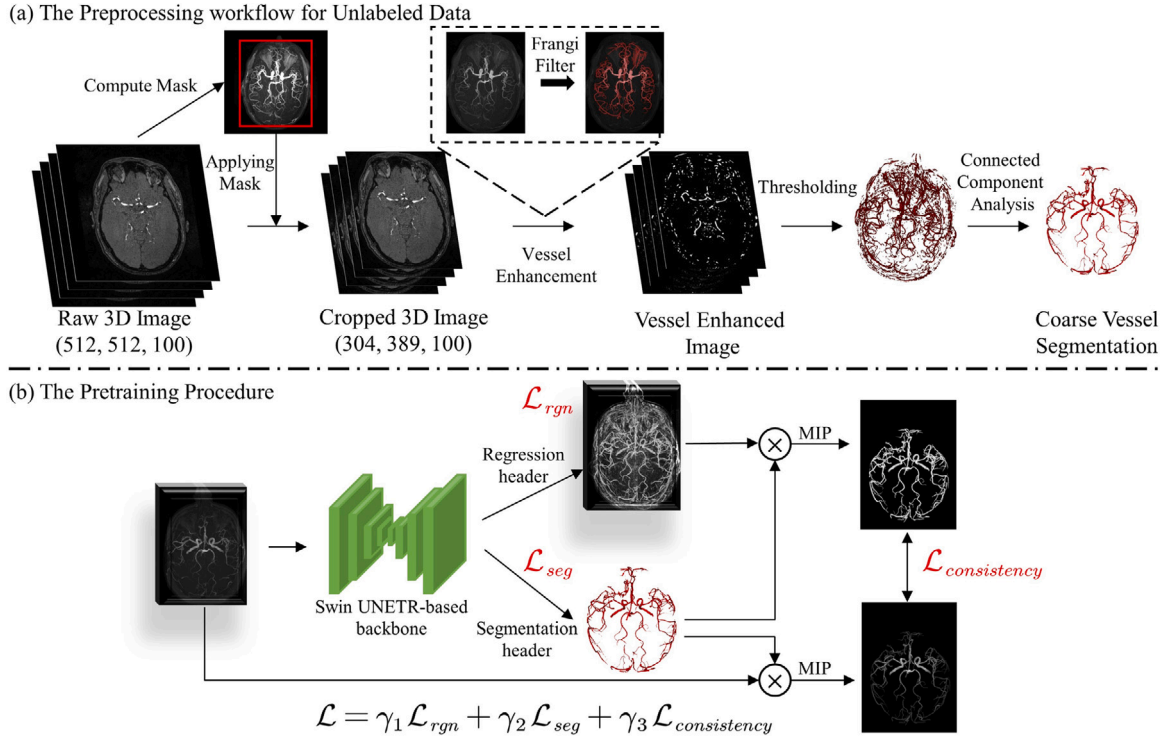


Fig. 2. (a) The illustration of the proposed preprocessing workflow. (b) The overall framework of our proposed pretraining procedure with multi-task learning.

(Wang et al., 2023; Yu et al., 2023; Shi et al., 2023a; Liao et al., 2023; Shi et al., 2023b). For example, Chen et al. introduced the Generative Consistency for Semi-Supervised (GCS) model, which improves cerebrovascular segmentation by calculating the consistency of perturbed data (Chen et al., 2022a). Similarly, Shen et al. proposed a novel framework (PoissonSeg) that applies Poisson learning to SemiSL for medical imaging (Shen et al., 2021). However, in real-world scenarios, unlabeled data are significantly larger and often exhibit greater heterogeneity, leading to a data heterogeneity problem between unlabeled and labeled data. This heterogeneity is known to reduce the effectiveness of SemiSL methods (Xu et al., 2024; Jiao et al., 2023). Additionally, one study (Oliver et al., 2018) shows that incorporating unlabeled data from inconsistent distributions can degrade performance compared to not using it at all. Current SemiSL methods, including the aforementioned GCS, PoissonSeg, and baseline models (Luo et al., 2021; Bai et al., 2023), primarily rely on relatively small amounts of unlabeled data that are assumed to share the same distribution as the labeled data. Consequently, their performance under real-world conditions remains unclear. This highlights the necessity for further validation of these methods in settings that better reflect the complexities of real-world data.

SSL methods have also shown significant efficacy in medical imaging research (Zhang et al., 2023; Krishnan et al., 2022; Yu et al., 2022). For instance, Tang et al. implemented a multitask pretraining strategy using a 3D Swin transformer, which has been notably successful in segmenting multiple organs (Tang et al., 2022). Dang et al. proposed a novel weakly supervised learning and SSL framework that achieves performance comparable to fully supervised methods (Dang et al., 2022). However, the pretraining tasks in current SSL methods are often not well aligned with downstream tasks, limiting their ability to enhance performance on one singular task. Therefore, incorporating domain-specific knowledge into pretraining tasks and evaluating model performance on datasets that more closely reflect real-world scenarios is worth further exploration.

3. Methods

The preprocessing workflow and pretraining procedures are illustrated in Fig. 2a and b, respectively. Initially, the preprocessing workflow reduces the image size to decrease computational costs. It then processes the unlabeled TOF-MRA data to generate vessel enhanced image (VEI) and coarse vessel segmentation (CVS). In the pretraining phase, these VEIs and CVS serve as targets for regression and segmentation tasks, respectively.

3.1. Preprocessing method for unlabeled data

The pretraining process consists of three steps: image size cropping, vessel-structure enhancement, and coarse segmentation.

3.1.1. Image size cropping

The TOF-MRA images often contain a significant number of background pixels, which have no relevance to the vessel segmentation task. Removing these background pixels provides a key advantage by reducing computational costs, especially when dealing with a large-scale unlabeled dataset. Previous approaches have utilized non-zero masks to crop raw images (Isensee et al., 2021). However, background pixels do not always possess intensity values of zero. In this study, we propose a novel method to calculate the cropping mask based on the variation of pixel intensities along the Z-axis (slice direction).

For a given single image, denoted as I_i with dimensions (H, W, D) , we compute three projection images along the Z-axis: Average Intensity Projection (AIP), Maximum Intensity Projection (MIP), and Intensity Variation Map (IVM). The calculations are as follows:

$$AIP_i = \text{Average}(I_i) \quad (1)$$

$$MIP_i = \text{Maximum}(I_i)$$

$$IVM_i = \text{STD}(I_i)$$

To generate the cropping mask, threshold segmentation is employed. We empirically set a cropping threshold β at 0.35 (The impact of the threshold β on the efficiency of the cropping process is analyzed

in the results section.). Using the MIP as an illustrative example, a mask is created by retaining only the top β percentage of pixels based on intensity in the MIP image. The resulting masks from the AIP, MIP, and IVM are subsequently merged using a bitwise AND operation. Furthermore, smaller regions comprising fewer than 200 pixels are eliminated to maintain focus on main vessel regions. The finalized square cropping mask is then consistently applied to each slice of the I_i .

3.1.2. Frangi filtering

For 3D TOF-MRA images, the Frangi filter involves several steps to enhance the visibility of blood vessels. First, it computes the second-order mixed partial derivatives for each pixel using the Hessian matrix. The Hessian matrix is defined as:

$$H = \begin{bmatrix} \frac{\partial^2 I}{\partial x^2} & \frac{\partial^2 I}{\partial x \partial y} & \frac{\partial^2 I}{\partial x \partial z} \\ \frac{\partial^2 I}{\partial y \partial x} & \frac{\partial^2 I}{\partial y^2} & \frac{\partial^2 I}{\partial y \partial z} \\ \frac{\partial^2 I}{\partial z \partial x} & \frac{\partial^2 I}{\partial z \partial y} & \frac{\partial^2 I}{\partial z^2} \end{bmatrix} \quad (2)$$

where $\frac{\partial^2 I}{\partial x^2}$ represents the second partial derivative of the image intensity I with respect to the x coordinate, and similarly for the other partial derivatives. Next, the eigenvalues of the Hessian matrix, denoted as λ_1 , λ_2 , and λ_3 , are calculated. These eigenvalues provide information about the local structure of the image. Specifically, the magnitude and orientation of the eigenvalues help identify pixels that belong to blood vessels.

In general, larger positive eigenvalues correspond to the central region of vessels, smaller positive eigenvalues represent the vessel edges, while negative eigenvalues typically indicate irrelevant structures and backgrounds. By analyzing the eigenvalues, pixels with vascular features are selected, and the VEI is obtained at this step.

3.1.3. Coarse segmentation

To achieve a coarse segmentation of the blood vessels, we employ a combination of thresholding and connected component analysis. Initially, a segmentation threshold α is empirically set at 0.95. This threshold is utilized to create a vessel mask by excluding the lowest α proportion of pixels in the VEIs, effectively isolating vessel pixels from the background and other anatomical structures. Subsequently, connected component analysis is conducted on the vessel mask. We retain the largest k connected regions, with k typically ranging between 3 and 5 based on empirical selection, to represent the coarse segmentation of the blood vessels. k is set as 3 in this study.

3.2. Pretraining task and loss function

Our proposed pretraining process contains three tasks: regression learning, coarse vessel segmentation learning, and consistency learning in the MIP projection. Let function $f(\cdot)$ be the backbone neural network, and V_i and S_i are the VEI and CVS for subject i . $f(\cdot)$ is used to encode the raw images, and one regression header $Rgn(\cdot)$ and $Seg(\cdot)$ segmentation header are used to predict V_i and S_i , respectively. It can be formulated as:

$$\hat{V}_i = Rgn(f(I_i)) \quad (3)$$

$$\hat{S}_i = Seg(f(I_i))$$

where $Rgn(\cdot)$ and $Seg(\cdot)$ are convolution operation with kernel size 1. $\mathcal{L}_1$ is used to compute regression loss. Dice loss and Binary Cross Entropy are used to compute segmentation loss. These can be formulated as:

$$\mathcal{L}_{rgn} = \|\hat{V}_i - V_i\|_1 \quad (4)$$

$$\mathcal{L}_{seg} = \text{DiceLoss}(\hat{S}_i - S_i) + \text{BCELoss}(\hat{S}_i - S_i) \quad (5)$$

In this study, we do not conduct skull stripping for the preprocessed data. In such a case, the Frangi filter will also enhance the skull

pixels, while the CVS will not contain the skull pixels after connected component analysis. In order to focus more on learning the vascular structures, we incorporate an extra consistency loss by calculating the $\mathcal{L}_1$ loss between $\hat{S}_i \odot \hat{V}_i$ and $\hat{S}_i \odot I_i$ after applying maximum intensity projection along Z -axis. It can be formulated as:

$$\text{MIP}_{VS} = \text{Maximum}(\hat{S}_i \odot \hat{V}_i) \quad (6)$$

$$\text{MIP}_{IS} = \text{Maximum}(\hat{S}_i \odot I_i)$$

$$\mathcal{L}_{consistency} = \|\text{MIP}_{VS} - \text{MIP}_{IS}\|_1$$

where $\hat{S}_i \odot \hat{V}_i$ and $\hat{S}_i \odot I_i$ have almost removed the skull pixels. Meanwhile, the MIP images of TOF-MRA are commonly used for the initial assessment of vascular morphology. The consistency loss will help the model focus more on learning the vascular structures.

Finally, the total loss function $\mathcal{L}$ will be:

$$\mathcal{L} = \gamma_1 \mathcal{L}_{rgn} + \gamma_2 \mathcal{L}_{seg} + \gamma_3 \mathcal{L}_{consistency} \quad (7)$$

where γ_1 , γ_2 and γ_3 are the loss weight. We empirically set $\gamma_1 = 0.4$, $\gamma_2 = 0.4$ and $\gamma_3 = 0.2$ in this study.

3.3. Backbone

We select the Swin UNETR (Hatamizadeh et al., 2021) as the backbone in this study. Different from the vanilla model, successive convolutional layers are used in the initial embedding layers (Zhou et al., 2021a). This setting enhances the capture of spatial location information, a feature that is potentially beneficial to the segmentation of cerebrovascular structures with sparse distribution. Please note that we also examined the impact of our pretraining strategy on other backbone structures in the result section.

4. Experimental setup

4.1. Datasets

Unlabeled Datasets for Pretraining: We collect a large unlabeled TOF-MRA consisting of 1510 subjects from 5 public datasets:

- **IXI**¹: This dataset contains 570 subjects. 45 subjects with vessel masks (Chen et al., 2022b) are used for evaluation dataset. The rest 525 subjects are used for pretraining. The image size of most data (497 subjects) is (512, 512, 100). The image size of 27 subjects is (1024, 1024, 92) and one subject is (1024, 1024, 91). The spacing is (0.4688, 0.4688, 0.8000).
- **OASIS**²: This dataset contains 525 subjects (LaMontagne et al., 2019). The spacing and image size are not identical among the subjects. The average image size is (583, 765, 216). The spacing of most data (337 subjects) is (0.2995, 0.2995, 0.6000), and the average spacing is (0.2977, 0.2977, 0.5933).
- **TubeTK**³: This dataset contains 109 subjects. 42 subjects with vessel masks are used for the evaluation dataset. The rest 67 subjects are used for pretraining. The image size is (448, 448, 128) and the spacing is (0.5134, 0.5134, 0.8000).
- **ADAM**⁴: This dataset contains 113 subjects. Each subject contains one original MRA-TOF image and one bias field corrected MRA-TOF image. Only the original data are used in this study. The average image size is (556, 556, 131). The spacing is various and the average spacing is (0.3524, 0.3524, 0.5447).

¹ <http://brain-development.org/ixi-dataset/>.

² <https://www.oasis-brains.org>.

³ <https://public.kitware.com/Wiki/TubeTK/Data>.

⁴ <https://adam.isi.uu.nl/data/>.

Table 1

The cerebrovascular segmentation results on TubeTK-42, Brains, IXI-45 and EDEN datasets. The best and second-best results are shown in bold and underlined, respectively. The “Preprocessing” method provides coarse segmentation directly from the preprocessing procedure.

Datasets	TubeTK-42			Brains			IXI-45			EDEN		
Method	Dice	clDice	HD95	Dice	clDice	HD95	Dice	clDice	HD95	Dice	clDice	HD95
UNet (MICCAI'15)	72.98	81.33	9.09	73.77	75.86	3.72	79.82	75.23	10.25	79.79	80.62	9.89
DTC (AAAI'21)	69.31	76.35	25.60	64.36	64.86	34.72	70.44	63.48	63.31	71.65	65.65	91.93
GCS (TMI'22)	18.11	33.40	22.50	10.37	9.28	24.92	25.05	21.32	55.10	61.36	67.85	28.42
GBDL (CVPR'22)	36.40	55.67	16.16	39.79	52.84	63.71	12.46	20.37	42.43	51.78	64.48	11.41
SSNet (MICCAI'22)	65.15	71.84	28.65	64.22	63.88	35.04	73.55	65.14	52.05	69.72	64.26	96.65
BCP (CVPR'23)	73.36	81.95	8.33	77.25	80.31	2.88	81.37	80.64	6.17	86.44	<u>89.41</u>	2.71
PCRL (ICCV'21)	72.90	81.92	<u>7.82</u>	75.94	79.09	<u>2.74</u>	83.16	83.69	4.05	86.15	88.71	2.91
MAE (CVPR'22)	72.39	79.37	11.82	77.56	79.70	3.49	83.21	80.14	10.04	84.62	85.90	6.73
Swin UNETR (CVPR'22)	74.66	82.18	9.71	77.76	80.40	3.30	80.64	79.01	6.37	81.23	81.87	12.90
UniMiSS (ECCV'22)	70.54	<u>82.59</u>	9.03	77.63	79.62	3.03	85.45	83.49	3.83	<u>88.02</u>	88.51	2.91
GVSL (CVPR'23)	74.27	<u>82.57</u>	8.30	<u>77.87</u>	79.68	3.16	<u>85.92</u>	<u>84.82</u>	<u>3.35</u>	85.15	89.39	<u>2.61</u>
FFPN (Ours)	77.53	85.71	6.69	80.47	83.28	2.19	87.24	86.25	2.75	90.75	91.53	1.24
Preprocessing	29.64	31.27	71.29	31.73	37.01	38.55	43.18	55.50	38.56	68.48	69.43	46.18

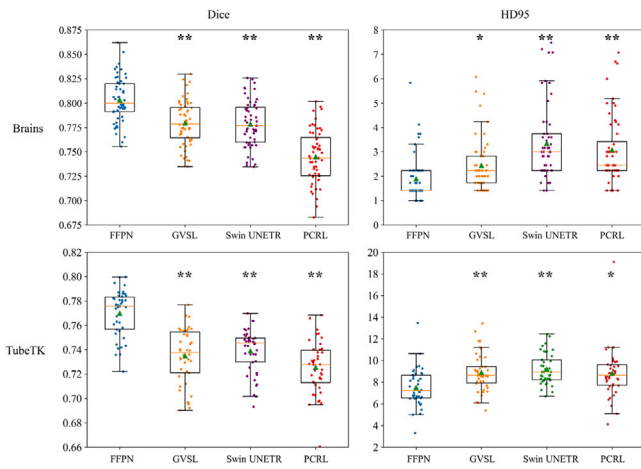


Fig. 3. Statistical validation on Brains dataset by using the T-test. ‘*’ and ‘**’ present $p < 0.05$ and $p < 0.001$, respectively.

- **BrainAneurysm⁵**: This dataset contains 284 subjects (Di Noto et al., 2023), of which 127 are healthy controls and 157 are patients with brain aneurysms. Four subjects (sub-200, sub-235, sub-315 and sub-450) are removed due to the incomplete image data. The average spacing is (0.4021, 0.4021, 0.6613), and the average image size is (466, 546, 147).

Labeled Datasets for Downstream Evaluation: After the pretraining procedure, four datasets with manually annotated vessel masks are used to evaluate their effectiveness. Please note that the pretraining dataset and evaluation dataset are completely isolated.

- **TubeTK-42:** The full TubeTK dataset contains 109 subjects, with 67 unlabeled subjects used for pretraining and 42 labeled subjects used for fine-tuning validation. The spacing is (0.5134, 0.5134, 0.8000) and the size of the images is (448, 448, 128). 33 subjects were utilized for fine-tuning the model, while 9 subjects were allocated for the testing phase.
- **Brains:** This dataset contains 56 subjects. The size of the images is not identical, with average sizes of (286, 320, 204). The spacing is (0.6188, 0.6188, 0.6200). 44 subjects were utilized for fine-tuning the model, while 12 subjects were allocated for the testing phase.

- **IXI-45:** This dataset is provided by Chen et al. (2022b). It contains 45 subjects from the whole IXI dataset (570 subjects). The image size is (1024, 1024, 92). The provided images have been processed and therefore do not contain spacing information. The original spacing of the IXI dataset is (0.2637, 0.2637, 0.8000). 36 subjects were utilized for fine-tuning the model, while 9 subjects were allocated for the testing phase.
- **EDEN:** This dataset contains 15 subjects. Unlike the previous dataset, we also provide the blood vessels on the skull in this dataset. The spacing is (0.2976, 0.2976, 0.4500) and the image shape is (672, 672, 210). 12 subjects were utilized for fine-tuning the model, while 3 subjects were allocated for the testing phase.

4.2. Evaluation metrics

Three metrics are used to assess the cerebrovascular segmentation performance, including the Dice similarity coefficient (Dice), Hausdorff Distance 95% (HD95), and centerline Dice coefficient (clDice) (Shit et al., 2021). The Dice and HD95 metrics primarily focus on the overlap degree and shape differences in segmentation results, clDice takes into account the topological structure of the segmented regions, making it more suitable for evaluating tubular structures.

4.3. Baseline models and implementation

We compare the performance of 3D cerebrovascular segmentation between the proposed FFPN model and existing methods using unlabeled data, including the SemiSL methods BCP (Bai et al., 2023), GBDL (Wang and Lukasiewicz, 2022), GCS (Chen et al., 2022a), SSNet (Wu et al., 2022), DTC (Luo et al., 2021) and the SSL methods MAE (He et al., 2022), PCRL (Zhou et al., 2021b), UniMiSS (Xie et al., 2022), Swin UNETR (Tang et al., 2022), GVSL (He et al., 2023). Besides, we also include the basic UNet (Ronneberger et al., 2015) model that does not utilize the unlabeled data to provide a comparison.

For our proposed FFPN model, the encoder has 4 stages which comprise of [2, 2, 6, 2] transformer blocks at each stage. The initial patch size is 2 and the feature size is 24. In pretraining task learning, the model is trained using AdamW optimizer for 200k iterations. Warm-up cosine scheduler of 20k iterations is used. The cropping patch size is (160, 160, 64) and the batch size is 2. The learning rate is 5×10^{-3} . In the fine-tuning procedure, the model is trained using AdamW optimizer for 5k iterations. The learning rate is 5×10^{-3} and the cropping patch size is (192, 192, 96).

⁵ <https://openneuro.org/datasets/ds003949/versions/1.0.1>.

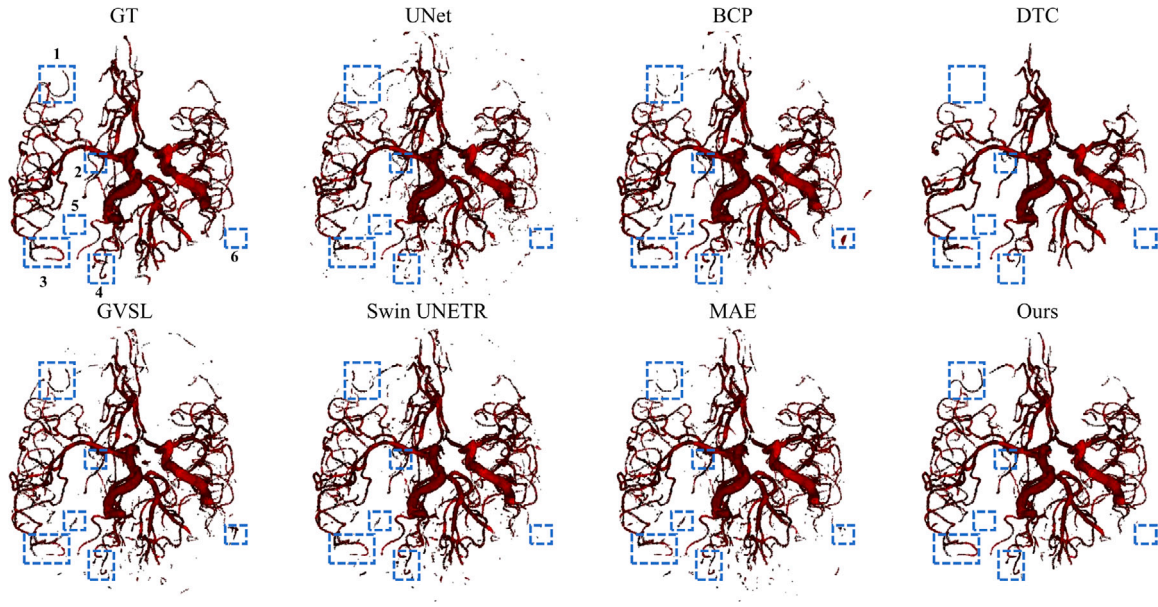


Fig. 4. The visualization of the segmented vessel of our method and the competing baseline models from the 3D view.

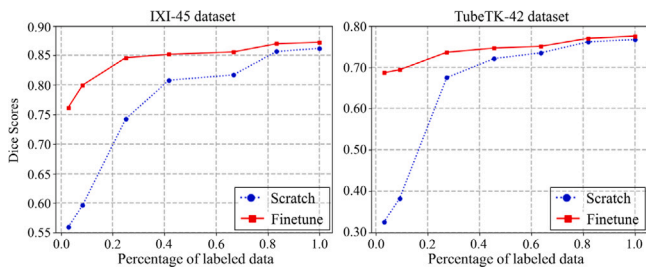


Fig. 5. Variation of the effect of the pretraining strategy with the labeled data percentage.

5. Experimental results

5.1. Comparison with state-of-the-art methods

The cerebrovascular segmentation results for the four datasets are shown in Table 1. The FFPN model achieves superior performance compared with other baseline models on the four datasets in terms of the three metrics. In TubeTK-42 and Brains datasets, the cIDice scores of our proposed method are 85.71 and 83.28, with an improvement of approximately 3% over the second model. In IXI-45 and EDEN datasets, our proposed method also achieves an improvement of around 2% compared with the second-best model.

To further demonstrate the superiority of the FFPN model, we conducted a comprehensive statistical analysis, the results of which are illustrated in Fig. 3. Specifically, we performed an additional five-fold cross-validation on the Brains and TubeTK-42 datasets using FFPN and baseline models with comparable performance, including GVSL, Swin UNETR, and PCRL. Subsequently, a paired T-test was executed to compare the performance of FFPN against each baseline method. The results indicate that the improvements rendered by FFPN are statistically significant across three metrics (the outcome for the metric cIDice aligns closely with that of Dice). This analysis robustly demonstrates the enhanced performance capabilities of FFPN.

We provide a 3D visualization result to support an intuitive evaluation (see Fig. 4). Compared with other methods, the segmentation

Table 2

The effectiveness of different loss functions.

Dataset	TubeTK-42			EDEN		
Loss	Dice	cIDice	HD95	Dice	cIDice	HD95
Scratch	72.08	80.16	11.46	88.87	90.53	2.13
$\mathcal{L}_{rgn}$	72.34	81.11	9.62	89.37	90.68	2.15
$\mathcal{L}_{seg}$	73.46	81.58	9.10	89.76	90.97	1.80
$\mathcal{L}_{consistency}$	72.15	80.78	10.13	88.96	90.83	1.94
$\mathcal{L}_{rgn} + \mathcal{L}_{seg}$	74.18	82.17	8.14	90.17	91.24	1.41
All	74.64	82.94	7.57	90.75	91.53	1.24

produced by FFPN reveals marked superiority in achieving continuity for the vessel structure, particularly noticeable at locations 1, 2, 3, and 4. Furthermore, it is worth highlighting that the FFPN-enabled segmentation leads to a reduction in false positive points (e.g., locations 5 and 6). Additional visualization cases are available in the supplementary material.

5.2. Influence of the percentage of manual labeled data

We then conduct a study to evaluate the effectiveness of our pre-training method using a reduced amount of manually labeled data. The results are presented in Fig. 5. The percentage of labeled data refers to the ratio of training data used in each experiment, compared to the amount utilized under a fully supervised scenario as detailed in Section 5.1. For example, in the context of the IXI-45 datasets, a 41.67% percentage corresponds to 15 cases being used for training as opposed to the 36 cases employed in a fully supervised approach.

It is observed that pretraining consistently improves the model's performance with different proportions of labeled data used. When using all available labeled data, our pretraining method achieves a 1.03% higher Dice score for the IXI-45 dataset and 0.83% higher Dice score for the TubeTK-42 dataset compared to a model trained from scratch. Additionally, with less than half of the labeled data, the pretrained model demonstrates an improvement of 4.39% for the IXI-45 dataset and 2.56% for the TubeTK-42 dataset. Furthermore, as the availability of labeled images decreases, the performance gap between the models trained with and without pretraining increases.

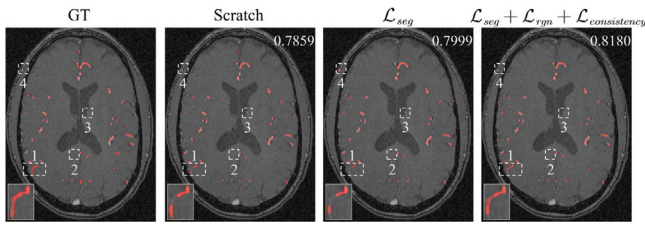


Fig. 6. Illustration of the effectiveness of pretraining loss functions from the slice view. The number indicates the cDice metric.

5.3. Effectiveness of pretraining loss function

This experiment was designed to evaluate the efficacy of our proposed loss function within the pretraining procedure. The model was pretrained using different loss functions, and their effectiveness was subsequently assessed during the fine-tuning phase. We utilized the TubeTK-42 dataset, which comprises 45.45% labeled data (equivalent to 15 subjects), and the EDEN dataset to demonstrate this, with the results detailed in Table 2.

On the TubeTK-42 dataset, the best performance in single-task pretraining was observed in segmentation learning, yielding Dice, cDice, and HD95 values of 73.46, 81.58, and 9.10, respectively. Regression learning displayed similar outcomes, whereas consistency learning did not markedly enhance the model's performance. The combination of regression and segmentation learning loss functions led to further improvements in performance. Incorporating all three learning tasks—segmentation, regression, and consistency—resulted in the highest overall performance, achieving Dice, cDice, and HD95 scores of 74.64, 82.94, and 7.57, respectively. A similar performance trend was noted in the EDEN dataset, underscoring the robustness and adaptability of our approach across different data sets.

We also present a visualization result from the slice view in Fig. 6. Incorporating pretraining with $\mathcal{L}_{seg}$ aids in reducing the occurrence of false positive points in the generated output (e.g., locations 2 and 3). However, it is worth noting that due to the coarse segmentation results potentially retaining some information from the skull, there may still be instances where certain structures on the skull are annotated (e.g., location 4). Overall, by incorporating all pretraining tasks, the model achieves improved segmentation, characterized by enhanced continuity of vessels (see location 1) and a reduced presence of false positive points.

5.4. Influence of model components for pretraining

Unlike conventional SSL methods, which generally transfer only the encoder parameters to downstream tasks, our proposed approach incorporates both the encoder and decoder parameters, along with the segmentation header. In this section, we investigate the effects of loading various components of the model on performance. We utilize the IXI-45 dataset, which comprises 41.67% labeled data (equivalent to 15 subjects), and the EDEN dataset to illustrate this impact. The corresponding results are detailed in Table 3.

On the IXI dataset, loading only the encoder results in a marginal performance enhancement, with a Dice score of 81.71, a cDice score of 79.90, and an HD95 value of 12.08. A comparable improvement is noted when the decoder alone is loaded. However, the performance sees a significant increase when both the encoder and decoder are loaded together, achieving a Dice score of 83.77, a cDice score of 81.57, and an HD95 value of 6.59. The optimum results on the IXI dataset are observed when all components, including the encoder, decoder, and segmentation header, are loaded, with a Dice score of 85.19, a cDice score of 83.09, and an HD95 value of 4.43. A similar performance trend is evident in the results for the EDEN dataset.

Table 3

The effectiveness of different model component pretraining parameters loaded.

Dataset	IXI-45			EDEN		
	Dice	cDice	HD95	Dice	cDice	HD95
Scratch	80.79	78.08	8.41	88.87	90.53	2.13
Encoder	81.71	79.90	12.08	89.92	90.91	1.82
Decoder	81.14	76.22	11.19	89.15	90.60	2.09
En. + Decoder	83.77	81.57	6.59	90.21	91.21	1.47
All	85.19	83.09	4.43	90.75	91.53	1.24

5.5. The effectiveness on different backbone structures

We assess the effectiveness of the proposed pretraining strategy on different backbone structures in this experiment. We focus on four fundamental medical segmentation backbones, which encompass both convolutional neural network (CNN)-based models (UNet (Ronneberger et al., 2015) and AttentionUNet (Oktay et al., 2018)) and Transformer-based models (UNETR (Hatamizadeh et al., 2022) and Swin UNETR (Hatamizadeh et al., 2021)). The corresponding results are summarized in Table 4.

For the EDEN dataset, we observe improvements when using our proposed pretraining learning compared to training from scratch across all backbone architectures. Specifically, for the UNet backbone, the Dice score increases from 79.79 to 84.26, resulting in a significant improvement of 4.47%. Similarly, the AttentionUNet and UNETR backbones show improvements of 9.76% and 4.74%, respectively. Even for the advanced Swin UNETR backbone, there is still a notable 1.88% improvement.

In the case of the IXI-45 dataset, we observe similar trends. Our proposed pretraining learning consistently leads to improvements compared to training from scratch. The UNet backbone shows an improvement of 5.41%, while the AttentionUNet, UNETR, and Swin UNETR backbones show improvements of 3.10%, 3.84%, and 1.03%, respectively.

5.6. The robustness of our pretraining method regarding the unlabeled dataset heterogeneity

We carried out this experiment to assess the robustness of our proposed method in relation to multi-source heterogeneous pretraining data, using the IXI dataset as an illustrative case.

In this experiment, we kept the total amount of unlabeled data constant at 525 subjects while varying the composition of this data. We tested three configurations: exclusively using the IXI dataset, exclusively using the OASIS dataset, and a mixed dataset comprising approximately half (262 cases) from both IXI and OASIS datasets. Subsequently, we evaluated the model's performance on the labeled IXI-45 dataset. For comparison, we used the best-performing semiSL method, BCP. The result can be seen in Fig. 7a. We also evaluate the FFPN performance across different labeled data ratios (Fig. 7b). The results indicate that the performance of the BCP method declines significantly as the proportion of homogeneous data decreases. In contrast, our proposed method exhibits consistently stable segmentation performance across all settings, regardless of the labeled data ratio. This outcome underscores the robustness of our pretraining strategy and it may not be necessary to augment the dataset with more labeled samples when dealing with heterogeneous data.

5.7. Analysis of the preprocessing procedure

Cropping Process: Our proposed cropping strategy can reduce the computation cost when processing such a large 3D MRA dataset. We take the IXI dataset as an example to show the effectiveness of the proposed cropping strategy. We compute the cropping rate (CR) for

Table 4

The effectiveness of our proposed pretraining method on various backbone structures. The number indicates the Dice scores.

Dataset	EDEN		
Backbone	Scratch	Fine-tune	Improvement
UNet	79.79	84.26	+4.47%
AttentionUNet	78.44	88.20	+9.76%
UNETR	83.32	88.06	+4.74%
Swin UNETR	88.87	90.75	+1.88%

Dataset	IXI-45		
Backbone	Scratch	Fine-tune	Improvement
UNet	79.82	85.24	+5.41%
AttentionUNet	81.52	84.62	+3.10%
UNETR	80.01	83.85	+3.84%
Swin UNETR	86.21	87.24	+1.03%

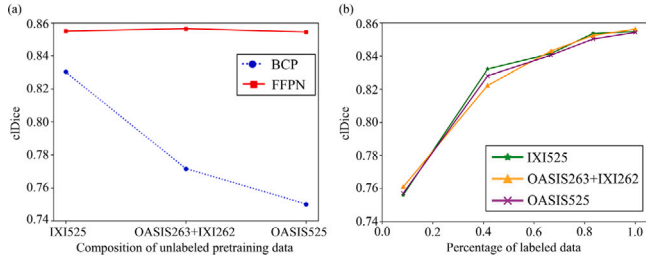


Fig. 7. (a) Performance comparison between FFPN and the best-performing SemiSL method, BCP, under varying compositions of the pretraining unlabeled dataset. The horizontal axis represents the composition of unlabeled data used during pretraining. (b) Robustness analysis of the proposed method with varying percentages of labeled data during the fine-tuning process.

each subject as the following formula:

$$CR = \frac{H \cdot W \cdot D - H' \cdot W' \cdot D'}{H \cdot W \cdot D} \quad (8)$$

where (H, W, D) is the original image size and (H', W', D') is the image size after cropping. Higher CR represents more background pixels being removed. The CR of the IXI dataset is shown in Fig. 8a. The average CR is 42.99%, and 80% of the samples subtracted nearly 40% (38.49%) of the background points.

Additionally, we examine the impact of the threshold parameter β on the cropping rate, with the findings illustrated in Fig. 8b. As β decreases, the resultant cropped image size diminishes, indicating that more background pixels are removed. Notably, when $\beta < 0.35$, the cropping results stabilize, suggesting an optimal threshold range for effective image processing. We also assess the robustness of this approach when handling low-quality, high-noise images. Specifically, four degradation types are considered: Gaussian noise, Salt-and-Pepper noise, downsampling in k-space, and intensity inhomogeneity (visualizations can be found in Supplementary Material Fig. 10). The results demonstrate that the proposed cropping strategy remains consistent across different conditions. Two representative data are shown in Fig. 8d.

Segmentation Threshold α : In this subsection, we analyze how the parameter α influences the classification of pixel points as blood vessels. α is varied within the range (0.9, 0.975), and the analysis is conducted using the preprocessed data. The downstream performance is assessed on the Brains dataset, with results presented in Fig. 8c. It is observed that a α value of 0.95 yields optimal results, achieving the best performance in terms of the cDice and HD95 metrics. Besides, we evaluate the robustness of the pretraining strategy under the four degradation types using $\alpha = 0.95$. The performance of the HD95 and cDice metrics is shown in Fig. 8f and g, respectively. The results indicate that Gaussian noise and downsampling cause only a slight decrease in model performance, whereas intensity inhomogeneity and Salt-and-Pepper noise have a relatively larger impact on the HD95 metric.

A possible explanation is that intensity inhomogeneity and Salt-and-Pepper noise significantly reduce the richness of the extracted vessel structures (see Fig. 11a in the Supplementary Material) in the unlabeled data processed by our proposed method. As a result, models fine-tuned on this degraded data may lead to a decline in recognizing vessel boundaries, thereby lowering performance on the HD95 metric. Despite this, the proposed method will be fine-tuned on non-degraded data and demonstrate relatively stable performance across various degradation scenarios (see Fig. 11b in the Supplementary Material).

Annotation Efficiency: Our proposed framework employs Frangi filter and SSL strategy to generate vessel segmentation, the effectiveness of which is validated through a comparative analysis with another weakly-supervised-based SSL method in Dang et al. (2022). Please note that the labeling procedure was completed by one Ph.D. student, so the experience level and speed may not match those of the original VesselCAPTCHA study. We first make a comparison in terms of the pseudo-segmentation generation from one case (see 9a). VesselCAPTCHA identifies whether a 2D patch contains a vessel structure and then uses K-means clustering to generate pseudo-segmentation. The initial segmentation produced by VesselCAPTCHA approximates the distribution of blood vessels but may not be sufficiently accurate. We then assess the performance of the SSL method on the EDEN dataset. We applied VesselCAPTCHA's annotation process to the training set and predicted the unseen testing cases. The results are shown in Fig. 9b and Table 5. After SSL training, VesselCAPTCHA achieves superior performance compared to UNet and the Frangi filter, even without full supervision. While VesselCAPTCHA offers significant advantages particularly when no labeled data is available, it still requires some manual identification. Thus, it may not be suitable for handling large-scale unlabeled datasets due to the manual input required, which can significantly hinder workflow efficiency. For instance, fully utilizing the 1510 unlabeled cases in this study with VesselCAPTCHA would require performing the vessel identification procedure for part of the total dataset, resulting in an excessively time-consuming process. In contrast, our proposed framework generates pseudo masks almost without the manual labor required, making it more effective for processing large-scale unlabeled datasets.

6. Discussion

To unleash the potential of unlabeled TOF-MRA data for 3D cerebrovascular segmentation, our approach incorporates the Frangi filter as a key element in the pretraining phase. The proposed FFPN model demonstrates enhanced robustness and superior segmentation performance, notably outshining other SemiSL and SSL methodologies.

In our preprocessing procedure, we do not perform the skull stripping step (Smith, 2002). Skull stripping is a basic process used by conventional cerebrovascular segmentation methods to remove the influence of the skull. The proposed cropping strategy aims to decrease the computation cost by removing the noisy background pixels. Besides, some end-to-end methods may omit this preprocessing step (Guo et al., 2024; Chen et al., 2024), particularly in general 3D vascular segmentation approaches (Han et al., 2024; Liu et al., 2023). These methods are also validated on TOF-MRA datasets and may omit skull-stripping to maintain consistency with other vascular datasets. Our method may inadvertently generate artifact vessel masks in skull regions, as illustrated in the upper right part of Fig. 9a (Dashed boxes). Integrating skull stripping can help reduce artifacts but also increases computational cost. The effectiveness of skull stripping within our proposed framework will be explored in future studies. Additional analysis on skull stripping can be found in the Supplementary Material.

Our proposed method is not limited by the backbone structure and has relatively high generalizability. More importantly, the proposed pretraining strategy may reduce the performance gap among the various backbone models. It is observed from Table 4 that the performance gap between Swin UNETR and UNet is around 6.5 before pretraining,

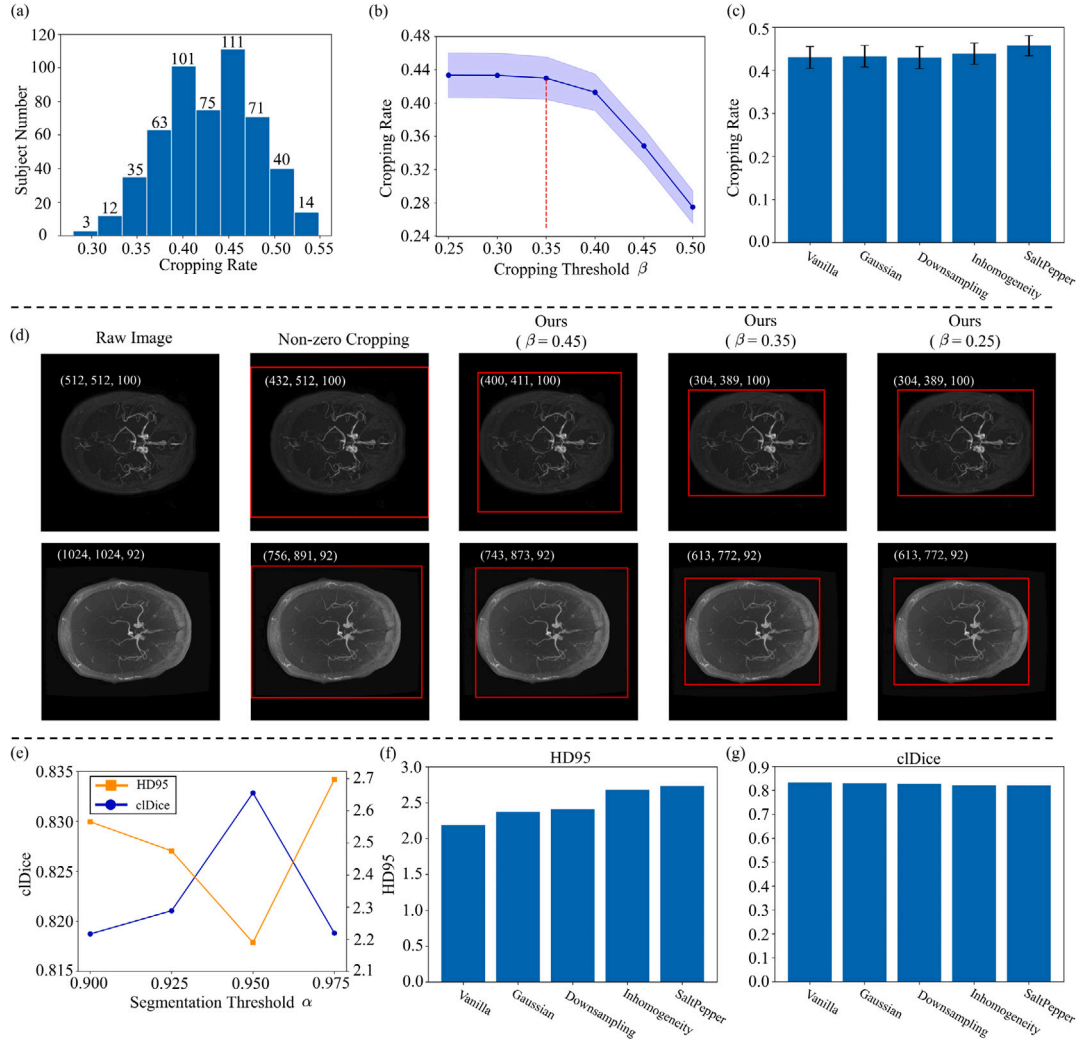


Fig. 8. The ablation analysis of the preprocessing procedure. (a) The distribution of the cropping rate in the IXI dataset. (b) The influence of cropping threshold β on the IXI dataset. (c) The impact of four types of degradations on the cropping strategy. (d) Two representative subjects for the cropping strategy. The number indicates the image size. (e) The influence of segmentation threshold α for the pretraining effectiveness on the Brains dataset. (f) and (g) The impact of four types of degradations on the pretraining procedure, evaluated using the HD95 and cDice metrics.

Table 5

The comparative analysis of label annotation efficiency on the EDEN dataset. Note that this table only considers the segmentation of intracranial vessels, so the metrics may differ slightly from those in the previous table for EDEN dataset.

Method	Extra unlabeled data	Label data	Labor time	Execution time	Dice	cDice	HD95
Frangi Filter	✗	✗	0.00	3.98 s	67.88	73.65	34.03
Vessel-CAPTCHA	✗	✗	86.11 min	8.47 s	83.80	87.78	2.70
UNet	✗	✓	–	–	77.38	82.71	5.54
FFPN	✓	✓	–	–	89.64	92.46	1.14

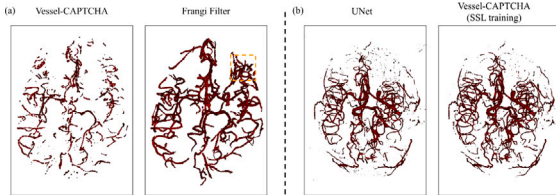


Fig. 9. (a) The comparison of annotation results from 3D views. Dashed boxes indicate artifacts in the skull. (b) The comparison of fully supervised UNet and weakly-supervised Vessel-CAPTCHA after SSL training.

while it becomes approximately 2.0 after using the proposed pretraining strategy. This indicates that the lightweight model (e.g., vanilla UNet) can also exhibit competitive performance after employing the proposed pretraining method, which holds significant implications for the use of lightweight networks in clinical settings.

One limitation of this study is that our method may not effectively use the other modal data that cannot fully reflect the vessel morphology (e.g., T1). The proposed pretraining method relies on the preprocessing workflow for cerebrovascular extraction, while other modal data might not include any vessel-related information. Although the proposed framework has been validated solely on TOF-MRA data, there is potential for adapting similar methodologies to other imaging modalities such as Digital Subtraction Angiography, Computed Tomographic Angiography, and Optical Coherence Tomography. While

direct application of the current method to these alternate modalities may not be straightforward, these modalities could still benefit from integrating the Frangi filter – a modality-independent technique for vascular enhancement – into the pretraining process. Subsequently, crafting modality-specific loss functions could improve model performance by aligning with the unique characteristics of each imaging type. Besides, our pretraining method can serve as an auxiliary training approach for a SSL model trained using other modal data or general large vision models, and help to construct a domain-specific model for cerebrovascular segmentation.

7. Conclusion

In this study, we introduce a Frangi filter-based pretraining network to effectively leverage the unlabeled TOF-MRA data. Our method capitalizes on the prior knowledge of tubular structures by incorporating it into the pretraining tasks and shows great superiority over other SemiSL and SSL methods on four cerebrovascular segmentation datasets. The ablation studies also show that our proposed method significantly improves model performance, particularly with a limited number of labeled data. Besides, our method is not restricted by model architecture and can enhance the performance of various backbone structures.

CRedit authorship contribution statement

Gen Shi: Writing – original draft, Visualization, Project administration, Formal analysis. **Hao Lu:** Writing – review & editing, Data curation. **Hui Hui:** Writing – review & editing, Data curation. **Jie Tian:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.media.2024.103442>.

Data availability

The code and data related to our research can be accessed via the following link: <https://github.com/shigen-StoneRoot/FFPN>. The repository contains the implementation and datasets associated with the FFPN method, and they are provided for public use and replication of our results.

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